

CS 301 — Data Structures and Algorithms

University of Example · Department of Computer Science

Fall 2026

Instructor: Dr. Lena Okonkwo

Office hours: Tuesdays and Thursdays, 2:00–3:30 p.m., Room 214

Course description

CS 301 introduces the data structures and algorithm-analysis habits used in later systems and software courses. Students implement the structures themselves, argue about running time in writing, and finish with an individual graph visualizer.

Weekly topics

Week 1: Asymptotic analysis. Big-O, big-Theta, and how to bound the running time of loops and recursive functions.

Week 2: Arrays and amortized analysis. Dynamic arrays, doubling strategies, and aggregate analysis.

Week 3: Linked lists and stacks. Singly and doubly linked lists, and stack-based evaluation.

Week 4: Queues and recursion. Circular queues, tail recursion, and the call stack.

Week 5: Trees and binary search trees. Traversals, insertion, and deletion.

Week 6: AVL trees and heaps. Rotations, the heap property, and priority queues.

Week 7: Hash tables. Chaining, open addressing, and choosing a hash function.

Week 8: Graphs and traversal. Representations, breadth-first search, and depth-first search.

Week 9: Shortest paths. Dijkstra's algorithm and when it does not apply.

Week 10: Sorting. Mergesort, quicksort, and the comparison lower bound.

Week 11: Dynamic programming. Overlapping subproblems, memoization, and tabulation.

Week 12: Review and project studio. Practice exams and graph visualizer critiques.

Assessments

Assignment 1: Complexity exercises. Due September 12, 2026. Weight 8%. Covers asymptotic analysis.

Assignment 2: Linked structures. Due September 26, 2026. Weight 8%. Covers linked lists and stacks.

Quiz 1: Foundations quiz. Due October 2, 2026. Weight 10%. Covers weeks 1 through 4.

Midterm Exam: Midterm exam. Due October 22, 2026. Weight 25%. Covers weeks 1 through 7.

Assignment 3: Hashing problems. Due November 6, 2026. Weight 8%. Covers hash tables.

Project: Graph visualizer. Due November 20, 2026. Weight 16%. Covers graphs and shortest paths.

Final Exam: Cumulative final exam. Due December 10, 2026. Weight 25%. Covers every weekly topic.

Policies

Late work loses 10% per day for four days and is not accepted after that. Exams are closed-book except for one handwritten sheet. The project is individual work. Office hours are the best place to debug a proof or a failing test.

Grading

Assignments are 24% combined, the quiz is 10%, the project is 16%, the midterm is 25%, and the final exam is 25%.